Logan Bolton

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Education

New York University, Courant Institute of Mathematical Sciences Sep. 2026 – Dec. 2027
M.S. in Computer Science

Auburn University May 2026
B.S. in Computer Science, Minor in Statistics *GPA: 3.82*

Experience

Machine Learning Intern May 2026 – August 2026
Los Alamos National Laboratory *Los Alamos, NM*

- Designed a physics-informed neural network architecture for rotation invariant molecular simulations, improving prediction accuracy by 9% over prior models, resulting in a NeurIPS 2026 workshop paper.
- Sped up model inference by 45% through identifying sequential bottlenecks and rewriting them as parallel GPU operations with custom Triton kernels.

Undergraduate Researcher - Machine Learning June 2024 – May 2026
Dr. Nguyen's Artificial Intelligence Lab & Dr. Weimar's Sport Biomechanics Lab *Auburn, AL*

- First-authored a paper in ACCV (5.6% oral acceptance rate) examining how and why models like ChatGPT fail on extremely simple visual tasks. **Cited 300+ times** and featured in technical reports by researchers from **OpenAI**, **Google DeepMind** and **Anthropic**.
- Collaborated with researchers from Adobe to determine what percentage of real-world Photoshop use cases could be automated by image-editing AI models, publishing the findings in a first-author paper at WACV 2026.
- Trained computer vision pose estimation models to track canine joint positions in videos for a DoD-supported interdisciplinary project studying long-term canine health.

Machine Learning Engineering Intern May 2025 – August 2025
Corvid Technologies *Huntsville, AL*

- Trained transformer-based models to predict radar cross section (RCS) responses for 3D objects in high fidelity physics simulation software, resulting in 27% less prediction error than previous in-house methods.
- Deployed models to company servers with a user friendly graphical interface built with Gradio to allow users to run and view the results of models without any technical knowledge.

Full Stack Web Developer Co-Op August 2022 – August 2024
Campus Web Solutions *Auburn, AL*

- Used C#, ASP.NET, JavaScript, and SQL Server to build a user lookup and permissions tool for staff, saving ~4 hours/week of manual work.
- Rebuilt common search queries using indexed views, reducing average response time from 6.2 seconds to 180ms.

Projects

Facebook Marketplace Deal Finder | *Next.js, PostgreSQL* | Link 2026

- Developed an automated pipeline that scrapes Facebook Marketplace listings and uses OpenAI's API to estimate fair market value, notifying the user of items priced 30%+ below predicted value.
- Built a full-stack application with a Next.js frontend, Python backend, and PostgreSQL database that displays Marketplace listings alongside AI-generated fair market value estimates.

Technical Skills

Languages: Python, SQL, JavaScript, C#, Java, HTML/CSS
Frameworks/Libraries: PyTorch, Pandas, Scikit-learn, Node.js, Matplotlib, ASP.NET
Tools: AWS, Azure, Docker, Linux, Git
Certifications: Azure Fundamentals (AZ-900) – Microsoft Certification in Cloud Computing